INCH-POUND
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DETAIL SPECIFICATION

HOISTS, CHAIN, PNEUMATIC, LOW HEADROOM, TROLLEY-TYPE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pneumatic, trolley-type chain hoists with traverse drive, having extreme low headroom and zero load eccentricity. Hoists include a lifting unit, a traversing unit, and a pendant unit.

1.2 Classification. Hoists are of the following types or capacities, as specified (see 6.2).

a. Type I - 2,000 pounds rated load

b. Type II - 4,000 pounds rated load

c. Type III - 6,000 pounds rated load

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490 - Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-DTL-24441 - Paint, Epoxy-Polyamide, General Specification for

MIL-PRF-24635 - Coating System, Weather-Resistant, Exterior Use

MIL-C-24707/5 - Castings, Ductile Iron and Austenitic Ductile Iron

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-889 - Galvanic Compatibility of Electrically Conductive Materials

MIL-STD-1472 - Human Engineering

MIL-STD-1474 - Noise Limits

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Copies of these documents are available online at https://quicksearch.dla.mil.)

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 6013 - Standard for Industrial Enclosed Gear Drives

AGMA 6034 - Practice for Enclosed Cylindrical Wormgear Speed Reducers and Gearmotors

(Copies of these documents are available online at www.agma.org.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B29.1 - Precision Power Transmission Roller Chains, Attachments, and Sprockets

(Copies of this document are available online at www.asme.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 10 - Near-White Metal Wet Abrasive Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM A48/A48M - Standard Specification for Gray Iron Castings

ASTM A143/A143M - Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement

ASTM B26/B26M - Standard Specification for Aluminum-Alloy Sand Castings

ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529 - Degrees of Protection Provided by Enclosures (IP Code)

(Copies of this document are available online at www.iec.ch.)

SAE INTERNATIONAL

SAE AMS-A-21180 - Aluminum-Alloy Castings, High Strength

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Materials. Materials shall be as specified herein. Components that are in or supported by the Federal supply system shall be used to the fullest extent possible. Materials shall possess the properties necessary to withstand vibration, shock, and other conditions specified herein. Materials for use in exposed locations subject to salt water spray shall be free from all defects and imperfections that may affect serviceability or appearance of the hoists.

3.2.1 Metals. Lightweight, high strength alloys shall be used. Metals used in the construction of the hoists for exposed locations shall be, to the greatest extent possible, selected from the following:

a. Steel alloys up to yield strength levels of 130 kilopounds per square inch (kpsi).
b. Aluminum alloy castings in accordance with SAE AMS-A-21180 or ASTM B26/B26M.
c. Corrosion-resisting steel alloys 316 and 316L.
d. Stainless steel alloy 17-4PH, heat treated to a maximum yield strength of 130 kpsi.
e. Nickel-copper alloys.
f. Nickel-iron alloys.

When specified (see 6.2), permission for the use of metals or structural materials other than those listed may be granted.

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3.2.2 Prohibited materials. The hoist shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

a. Asbestos. Asbestos shall not be used.

b. Cadmium. Cadmium shall not be used.

c. Beryllium. Beryllium shall not be used.

d. Pressed steel. Pressed steel shall not be used except as specified for a particular application (see 6.2).

e. Magnesium. Magnesium or magnesium-based alloys shall not be used. This does not include steel or aluminum alloys that contain less than 0.5 percent magnesium.

f. Brittle material. Brittle material (see 6.7.1) shall not be used, except as specified for a particular application (see 6.2).

g. Cast iron. Cast iron in any form shall not be used except where permitted by referenced specifications. The use of cast iron is limited to those alloys conforming to ASTM A48/A48M, class 35 or better. Cast nodular graphitic iron or malleable iron conforming to MIL-C-24707/5 is acceptable for machinery equipment components.

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.4 Dissimilar metals. Coupled dissimilar metals shall be protected against corrosion in a marine environment in accordance with MIL-STD-889.

3.3 Safety. Hoists shall meet the human safety interface requirements of MIL-STD-1472. Equipment design shall consider the following safety criteria at a minimum:

a. Avoiding or eliminating identified hazards by design selection or material selection.

b. Controlling and minimizing hazards to personnel, equipment, and materials that cannot be avoided or eliminated.

c. Isolating hazardous substances, parts, and operations from other activities, areas, personnel, and incompatible materials.

d. Incorporating fail-safe design features so that failures will not cause injury to personnel or damage to equipment.

e. Locating equipment parts so that access to them by personnel during operation, maintenance, repair, or adjustment shall not require exposure to hazards such as chemical burns, electrical shock, sharp edges or points, or toxic atmospheres.

f. Providing warning and caution notes in operations, assembly, maintenance, and repair instructions, and distinctive warnings of hazards on the equipment.

3.4 Performance and characteristics. After installation on a rail, hoists shall exhibit the characteristics in 3.4.1 through 3.4.3 when visually inspected in accordance with 4.5.3 and tested in accordance with 4.5.3.1.

3.4.1 Type I. Performance and characteristics for type I hoists shall be as follows:

a. Rated load – not less than 2,000 pounds.

b. Lifting speed – not less than 25 feet per minute (ft/min).

c. Lift range – unless otherwise specified (see 6.2), not less than 8 feet.

d. Weight – not greater than 185 pounds, excluding the chain and hook.

e. Width – not greater than 12 inches to either side of the trolley centerline.

f. Length – 26 inches (not including optional cushion bumpers).

g. Height – the distance from the underside of the track to the lifting surface of the retracted load hook: minimum practicable, but not greater than 6.5 inches; in addition, other hoist projections, excluding the chain basket, shall extend not greater than 15 inches below the underside of the track. No part of the hoist shall extend 4 inches above the bottom of the trolley track.

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h. Rated air gauge pressure – 90 pounds per square inch (psi).

i. Air consumption – 90 to 100 standard cubic feet per minute (ft^3/min) at 90 psi.

j. Trolley track size – 5- by 3.000-inch by 10-pound-per-foot (lb/ft), 5- by 3.284-inch by 14.75-lb/ft, 6- by 3.330-inch by 12.5-lb/ft, and 6- by 3.565-inch by 17.25-lb/ft I-beam.

k. Trolley track radius – the trolley hoist shall negotiate S-curves of 26-inch radius.

l. Traversing speed – not greater than 40 ft/min with rated load and air pressure on the level and 10 ft/min on 10-degree incline.

3.4.2 Type II. Performance and characteristics for type II hoists shall be as follows:

a. Rated load – 4,000 pounds.

b. Lifting speed – not less than 16 ft/min.

c. Lift range – unless otherwise specified (see 6.2), not less than 8 feet.

d. Weight – not greater than 185 pounds for the hoist unit, excluding the chain and hook. Cumulative weight of the hoist, trolley, and hoist tractor units may total not greater than 225 pounds with 8 feet of lift chain and hook, if readily disconnected.

e. Width – not greater than 12 inches to either side of the trolley centerline.

f. Length – 26 inches (not including cushion bumpers).

g. Height – the distance from the underside of the track to the lifting surface of the retracted load hook: minimum practicable, but not greater than 6.5 inches; in addition, other hoist projections, excluding the chain basket, shall extend not greater than 15 inches below the underside of the track. No part of the hoist shall extend 4 inches above the bottom of the trolley track.

h. Rated air gauge pressure – 90 psi.

i. Air consumption – 90 to 100 ft^3/min at 90 psi.

j. Trolley track size – 5- by 3.000-inch by 10-lb/ft, 5- by 3.284-inch by 14.75-lb/ft, 6- by 3.330-inch by 12.5-lb/ft, 6- by 3.565-inch by 17.25-lb/ft, and 8- by 4.000-inch by 18.4-lb/ft I-beam.

k. Trolley track radius – the trolley hoist shall negotiate S-curves of 26-inch radius.

l. Traversing speed – not less than 40 ft/min with rated load and air pressure on the level and 10 ft/min on 10-degree incline.

3.4.3 Type III. Performance and characteristics for type III hoists shall be as follows:

a. Rated load – not less than 6,000 pounds.

b. Lifting speed – not less than 10 ft/min.

c. Lift range – unless otherwise specified (see 6.2), not less than 8 feet.

d. Weight – not greater than 285 pounds, excluding the chain and hook.

e. Width – not greater than 12.5 inches to either side of the trolley centerline.

f. Length – 31 inches (including optional cushion bumpers).

g. Height – the distance from the underside of the track to the lifting surface of the retracted load hook: minimum practicable, but not greater than 9.5 inches; in addition, other hoist projections, excluding the chain basket, shall extend not greater than 15 inches below the underside of the track. No part of the hoist shall extend 4 inches above the bottom of the trolley track.

h. Rated air gauge pressure – 90 psi.

i. Air consumption – 90 to 100 ft^3/min at 90 psi.

j. Trolley track size – 8- by 4.000-inch by 18.4-lb/ft I-beam, adaptable to 6- by 3.330-inch by 12.5-lb/ft and 6- by 3.565-inch by 17.25-lb/ft I-beam.

k. Trolley track radius – the trolley hoist shall negotiate S-curves of 36-inch radius.

l. Traversing speed – not less than 35 ft/min with rated load and air pressure on the level and 10 ft/min on 10-degree incline.

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3.5 Incline consideration. Hoisting and traversing brakes shall be provided that shall hold the rated load on an incline of ±15 degrees with the horizontal in any direction. The hoist shall operate on a ±10-degree incline with the horizontal in any direction, with rated load, at reduced speed, through the full lift range.

3.6 Design. Hoists shall be designed in accordance with the requirements specified herein.

3.6.1 Design philosophy. The design philosophy for the hoists is to achieve the performance, safety, maintainability, and reliability specified herein. The weight of the system shall be kept to a minimum (see 3.4.1, 3.4.2, and 3.4.3) while meeting the requirements in this section. At delivery, the net weight as well as the operating weight with fluids, where applicable, shall be indicated. The system shall be designed to reduce maintenance requirements (see 3.6.27 and 3.6.33.3).

3.6.2 Strength. Components shall have the strength to lift the required load (see 6.2) and operate satisfactorily in accordance with the requirements specified herein. Parts shall be as light and compact as practicable and consistent with the required strength and stiffness.

3.6.3 Moderate sea conditions. The hoist shall operate with rated load and at rated speed under moderate sea conditions and loading factors or motion conditions as specified (see 6.2).

3.6.4 Temperature. Hoists shall operate at temperatures of 0 through 130 °F for not less than 3,000 cycles without a failure.

3.6.5 Service life. Moving parts, except bearings, shall have a life in rated service of not less than 10,000 hours and shall operate for not less than 1,200 hours without adjustment or replacement of parts. Periodic adjustment of brakes due to lining or friction disc wear is expected.

3.6.6 Fail-safe operation. Hoists shall be constructed for fail-safe operation. Failure of the air source or manual drive mechanism shall not result in damage to the load, jeopardize the safety of personnel, or result in uncontrolled movement of the equipment and load. In the event of a loss of air, the hoist shall come to a controlled stop without damage. Restoration of air shall not result in automatic resumption of any equipment operation that may jeopardize the safety of the load, equipment, or personnel. Fail-safe requirements shall not be applied to static components, structural members, or other static parts of mechanisms.

3.6.7 Stress. Hoist mechanical components shall use the loading factors specified in 3.6.7.1. The combined stresses of mechanical components of the hoist, hoist tractor, and trolley shall not exceed 35 percent of the tensile yield strength of the material used for hoist operation at rated load under the 10-degree incline condition (see 3.5). The maximum combined stresses in structural and mechanical components under the 15-degree incline condition (see 3.5) shall be not greater than 70 percent of the yield strength of the material used when the hoist or hoist tractor is subjected to maximum torque or braking conditions and static and dynamic load tests (see 4.5.11.1 and 4.5.11.2).

3.6.7.1 Loading factor. Hoist components shall incorporate the following loading factors:

a. Dynamic loading based on inertial forces – 1.5 times rated load.

b. Static loading based on inertial forces – 2.0 times rated load.

3.6.8 Shock. When specified (see 6.2), the hoist, in the unloaded and stowed (not operating) condition, shall withstand the shock test for grade A or grade B equipment in accordance with MIL-DTL-901 (see 4.5.12) without permanent deformation or degradation of any operating functions. It will be permissible for the trolley hoist to drift along the track under shock conditions provided that no damage to the trolley hoist or brake results and the brake holds the hoist to the rail after shock. Drift shall be not greater than 1 inch.

3.6.9 Mechanical vibration. Unloaded hoists shall conform to the vibration requirements of MIL-STD-167-1, type I (see 4.5.13).

3.6.10 Noise. Each hoist shall be properly adjusted, balanced, and free from excessive noise during operation. The hoist airborne noise level shall be not greater than the maximum level of MIL-STD-1474, grade D when tested as specified in 4.5.14.

3.6.11 Nuclear lifting. Hoists to be used for nuclear lifting shall have a safety factor of 5 or greater for all hoist lifting components. Hoists to be used for nuclear lifting shall be as specified (see 6.2).

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3.6.12 General construction. Positive means shall be provided to prevent any component working loose. The hoist shall be complete in all respects, including operating controls and other equipment that may be necessary for operation. For hoists requiring repair parts, all wear parts shall be readily accessible for replacement. The hoist shall not be provided with an overload limiting device or load limiter.

3.6.13 Rotating shafts. Rotating shafts shall be supported in antifriction, lubricated, or self-lubricated bushings or bearings. Shaft bushings and bearings shall be enclosed against entry of dirt, dust, or foreign material. Rotating and sliding surfaces shall be lubricated.

3.6.14 Frame or housing. The housing shall be constructed of steel or aluminum alloy for maximum strength and minimum deflection. The frame or housing shall contain the hoist mechanisms, including gears, sprockets, load chain stowage, spring reel for chain drum or magazine, hoist and trolley brakes, protection of pneumatic controls and piping, air motors, and other operating components. No part of the hoist shall protrude below the motor except the pendant controls, chain bag, chain, load hook, or shackle, when applicable. Hoist or frame side tilt shall be kept to a minimum when operating on the trolley rail. In case of any inclination of the rail or effects of ship motion, no part of the hoist shall project above the top flange of the trolley rail. Means shall be provided to afford protection from damage due to bumping of two or more adjacent hoists or hoists on the same track. This protection shall be provided through use of bumpers or inherent frame or housing features.

3.6.15 Gears. Gears shall be spur, helical, or worm and wheel type manufactured in accordance with AGMA 6013 and AGMA 6034. Gears shall have an ingress protection (IP) rating of IP64, in accordance with IEC 60529, and be enclosed against dirt, dust, and water spray in a casing that permits ready access for inspection and cleaning.

3.6.16 Hoist drives. Hoist lift and hoist traverse units shall be driven by reversible air motors of the enclosed construction type, constructed to operate with air gauge pressure between 80 and 100 psi (dynamic) measured at the motor and 125 lb/in^2 (static) measured at the source. Motor brakes shall be furnished (see 3.6.23).

3.6.16.1 Hoist lift. The hoist lift pneumatic drive motor shall be coupled through a speed reducer or drive gear to the load chain sprocket.

3.6.16.2 Hoist traverse unit. The hoist traverse drive unit shall have positive traction. The motor shall be coupled through a speed reducer to a chain sprocket for use on a 1-inch pitch, single strand No. 80-1-RC-A standard roller chain, in accordance with ASME B29.1, welded to the bottom of the trolley track as shown on figure 1, or as specified (see 6.2).

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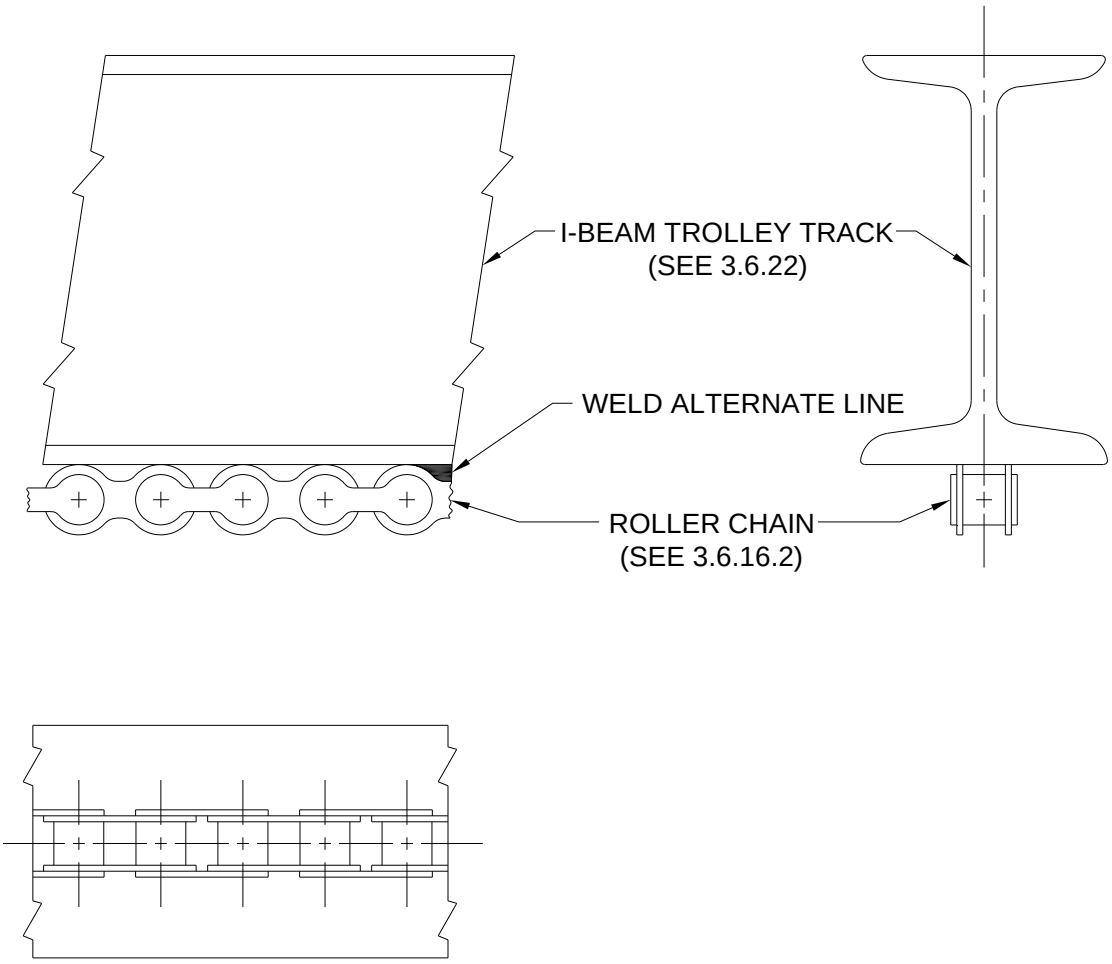


FIGURE 1. Sketch of trolley track.

3.6.17 Load chain. The hoist shall be double purchased. The load chain shall be carbon or alloy steel and shall be corrosion-resistant finished as specified in 3.6.17.1. The load chain shall be of sufficient hardness and strength to withstand the applicable tests specified in 4.5.3. The load chain shall be of the link chain type. Chain links shall be electric or forge welded. Each link shall be of uniform size and shape, free from scale and lamination at the welds, and shall seat properly in the hoist chain sheave pockets. The chain shall be free from any tendency to snarl. The ends of the load chain shall be securely attached to the hoist or provided with a means to prevent the end of the chain passing through the hoist and shall be of a configuration that will permit ready removal of the chain. The load chain shall provide a safety factor of not less than 5. This factor of safety shall be the ratio of chain breaking strength to the rated load applied to each purchase or 1,000 pounds for type I hoists, 2,000 pounds for type II hoists, and 3,000 pounds for type III hoists. A stripper shall be provided to free the chain from the sheave pockets on both sides of the sheave.

3.6.17.1 Load chain finish. The link load chain shall be protected to resist corrosion by electrodeposited zinc. The zinc coating shall conform to ASTM B633, type VI, class Fe/Zn 12. Processing shall be such that there is no impairment in free movement of any chain link after plating. The safeguarding against and procedure for detecting embrittlement of zinc coating shall be in accordance with ASTM A143/A143M.

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3.6.17.2 Load chain stowage. Hoist construction shall include a means for stowing the full length of the load chain in chain reels, bags, or baskets (see 6.2) when the hook is in the “up” position. The chain reels or baskets shall provide for stowage of chain for 8 feet of lift. The bags or baskets shall be used for stowage of chain for lifts greater than 8 feet. The chain reels shall maintain a relatively constant tension force to prevent chain slack between the load sprocket and the chain stowage. Construction of the load sprocket and load chain stowage shall provide a constant chain feed without binding or jamming in the chain guide, stowage, or hoist frame.

3.6.17.3 Load chain wheel, sprocket, and shaft. Load chain wheels or sprockets shall be provided with pockets accurately shaped to fit the links of the load chain. The load shaft shall be integral with or rigidly connected to the load chain wheel. Welding of the wheel to the shaft will not be permitted.

3.6.17.4 Chain guides. Chain guides that enclose the chain as necessary to ensure proper load sprocket action shall be provided. The function of the chain guides shall be to guide the load chain over the load sprocket at a uniform angle to prevent fouling and jamming of the load chain. The guides, if bolted on, shall not loosen under vibration.

3.6.17.5 Chain replacement. Hoist construction shall permit chain replacement by the use of hand tools available in the GSA Global Supply Catalog.

3.6.18 Limit stops. Upper and lower lift limit stops shall be provided.

3.6.19 Load attachments.

3.6.19.1 Load hooks. The hoist load hooks shall be of drop forged, heat-treated steel that shall withstand the test loads specified in Section 4. Hoist load hooks shall be as shown on figure 2, as specified (see 6.2), and shall be fitted with a swivel and a safety device, except type I, fixed eye, regular hooks. The safety device shall consist of a spring latch or swivel type closure to bridge the hook throat opening and shall be an integral part of the hoist hook. Spring latch type closures shall be hinged at the neck of the hook with the other end resting inside the tip of the hook. The swivel shall permit 360-degree rotation of the hoist load hook with full load without twisting the attached wire rope, link, or roller chain. Hoist load hook throat openings shall be in accordance with the dimensions shown in table I. The hook throat safety device of the suspension or upper hook shall withstand a pull against the hook throat safety device as specified (see 4.5.11). Hooks shall be clearly marked with manufacturer identification and allowable load or allowable load designator.

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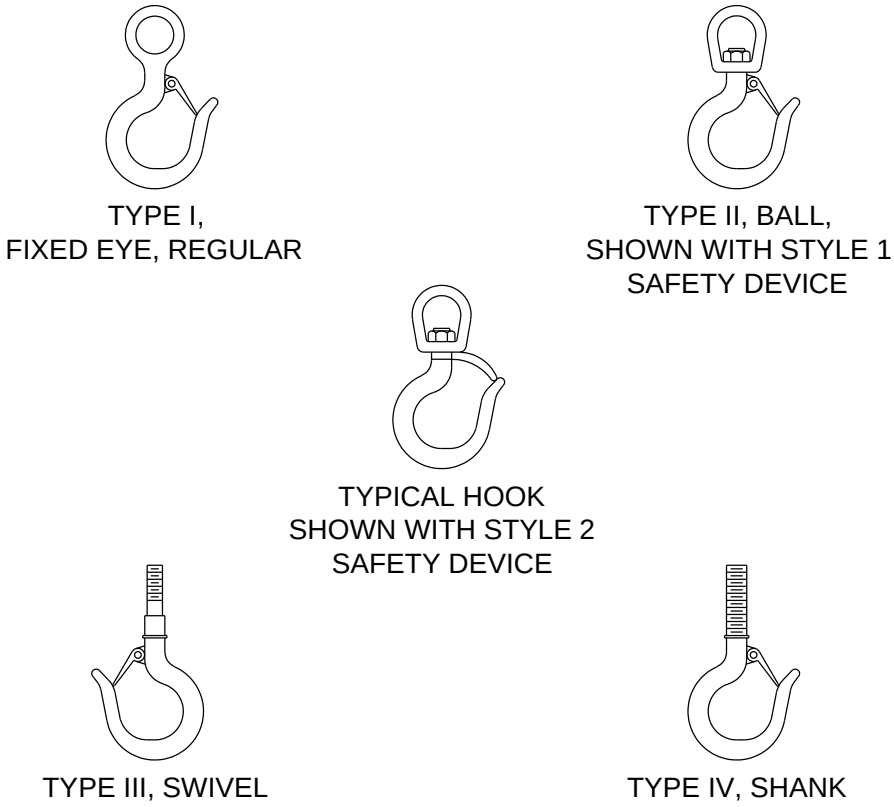


FIGURE 2. Illustrating types of hoist hooks (shown with safety devices, style 1 and style 2).

3.6.19.1.1 Load hook finish. Unless otherwise specified (see 6.2), hooks shall be zinc plated. Zinc plating shall be in accordance with ASTM B633, type VI, class Fe/Zn 12. The safeguarding against and procedure for detecting embrittlement of zinc coating shall be in accordance with ASTM A143/A143M. The hook throat safety device shall be constructed of non-corrosive material or treated for corrosion resistance.

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TABLE I. Hook throat openings.

Hoist rated load (pounds)	Hook throat opening (inches) (minimum)
1,000	0.750
2,000	0.906
3,000	1.0
4,000	1.125
5,000	1.125
6,000	1.5
7,500	1.375
10,000	1.625
11,000	2.0
13,000	2.063
15,000	2.063
17,000	2.063
20,000	2.25
25,000	2.25
30,000	2.75
40,000	3.0

3.6.19.2 Shackle. When required (see 6.2), a shackle with self-locking pin shall be substituted for the load hook to conserve head room. The shackle opening shall be a 1 1/16-inch wide by 1-inch clear opening above the upper edge of the pin. The shackle shall be provided with a thrust bearing and shall be free to rotate 360 degrees without binding under full load conditions.

3.6.20 Range of load hook. The hoist shall lift rated loads from any point within a 19-inch radius from an imaginary perpendicular under the chain sprockets to a horizontal plane 7 feet below the trolley track. The hoist shall lift rated loads at this offset range without binding or jamming of the load chain in the sprocket guide.

3.6.21 Bearings. Where practicable, rotating parts shall be mounted on antifriction bearings to suit the intended use and shall have a service life of not less than 10,000 hours. Bearings shall be selected by the hoist manufacturer with load ratings, including applicable safety factors, greater than the maximum bearing loads under extreme operating conditions specified herein. Exposed bearings shall be sealed or shielded. Bearings shall be of the permanently lubricated type, where practicable, or provided with fittings to add lubrication when required. When antifriction bearings are not used, bearings shall meet all requirements of antifriction bearings.

3.6.22 Trolley. The trolley hoist shall have not less than four wheels. Wheels shall have a lateral positioning feature enabling the trolley to operate, along a straight track, with a clearance of not less than 3/8 inch between the wheels and the web of the largest I-beam rail (see figures 3 and 4). Wheels shall be of ferrous material. The trolley shall operate on an I-beam sized as specified in 3.4.1 through 3.4.3. Flange widths are nominal and are subject to ±1/8-inch mill tolerance. The zero load eccentricity of the hoist shall be maintained when the wheels are positioned for each size rail.

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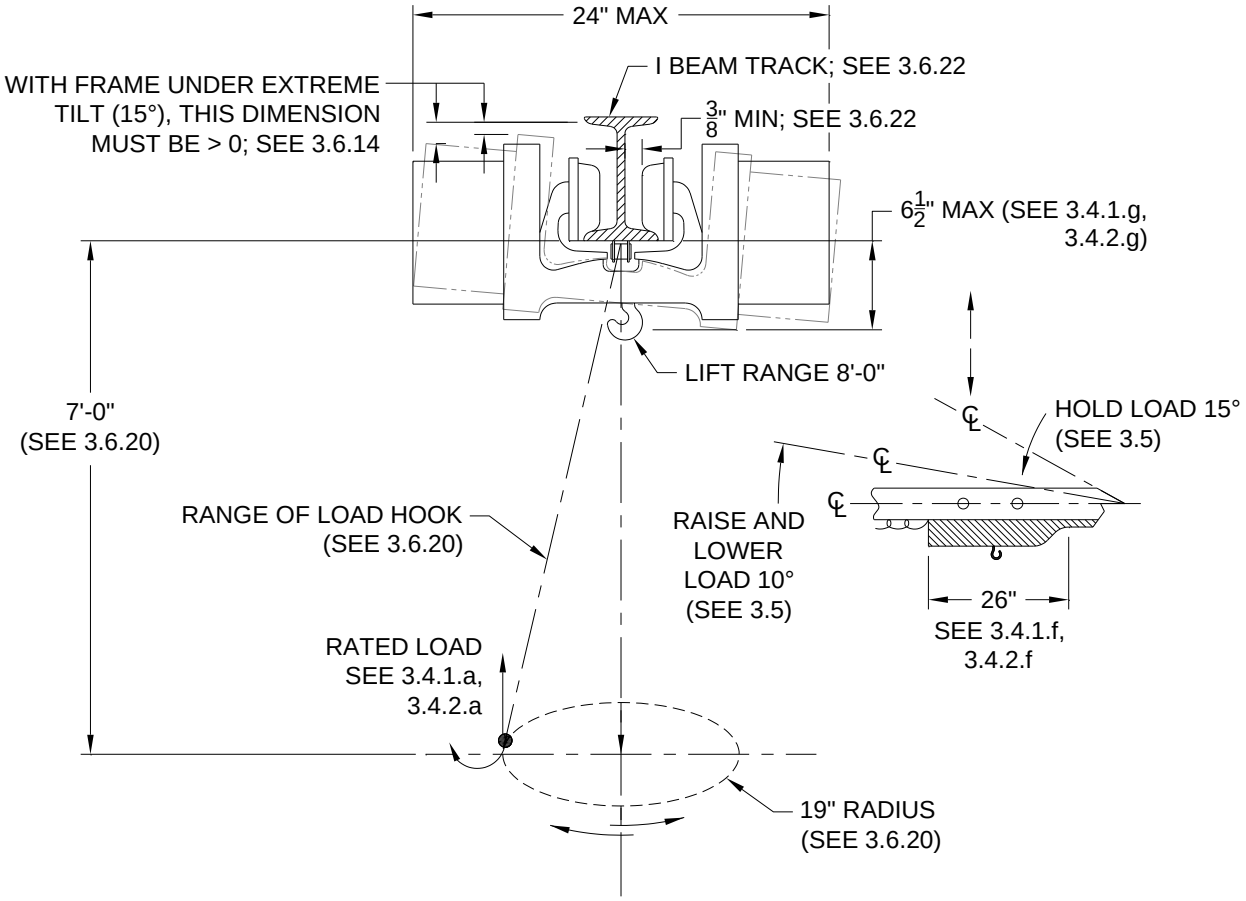


FIGURE 3. General requirements for type I and type II hoists.

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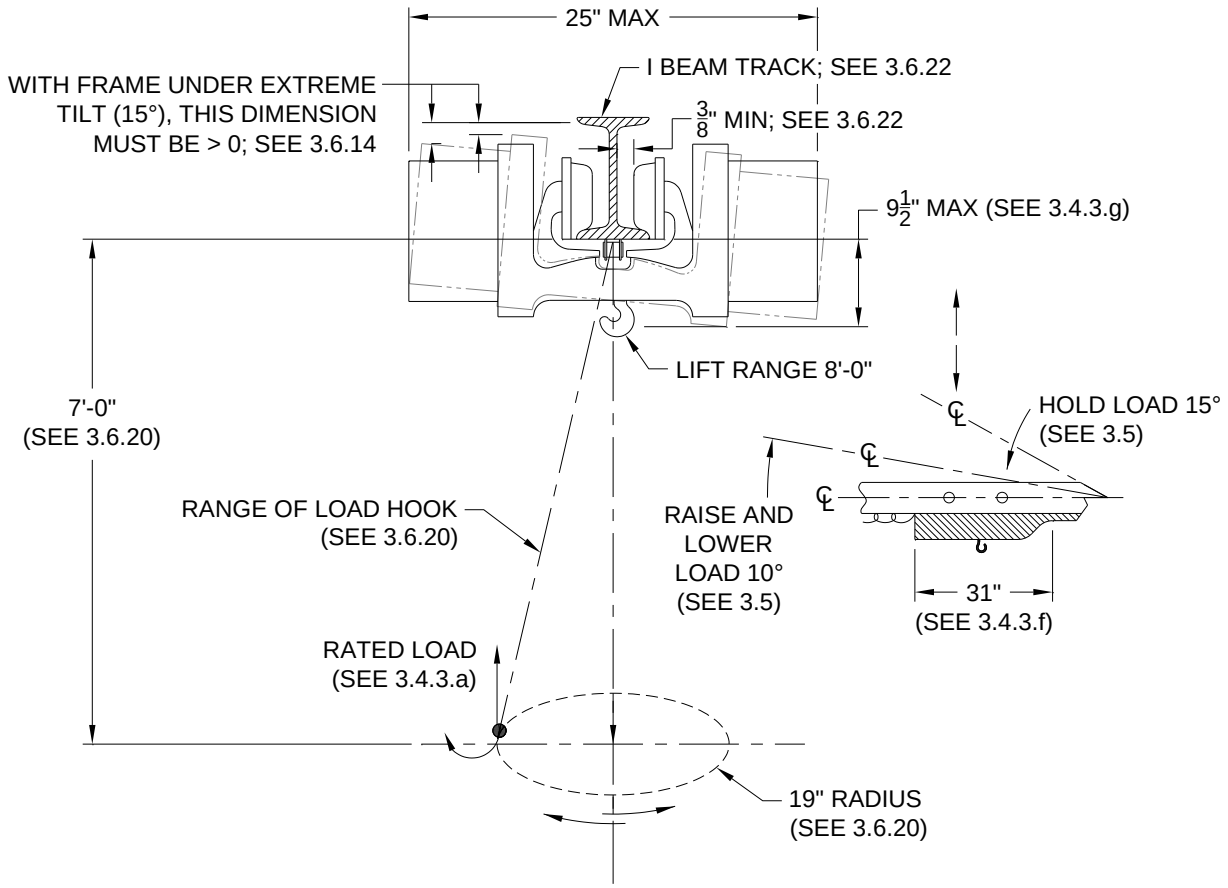


FIGURE 4. General requirements for type III hoists.

3.6.23 Brakes. The lifting and traversing power units shall be equipped with brakes coupled to the respective drive transmission. The brakes shall be the spring-loaded, automatic operating type constructed to actuate upon reduction of air pressure below the safe motor operating pressure. The brakes shall be equipped with a manual release for emergency operation. Manual release mechanisms shall be arranged so they can be operated without endangering the operator. Brakes shall hold the test loads without slipping from a stopped position and rated load on the hoist as specified in 3.5. Provisions shall be made to facilitate adjusting the brakes. If more than one hoist power unit is used, a load brake is required on each power unit, and provision shall be made to ensure that brakes will operate simultaneously.

3.6.24 Pendant unit. Controls shall be on a thumb-operated manifold suspended on a pendant of sufficient length to allow easy operation of the hoist for the required lift range (see 6.2). The pendant shall support a 200-pound load. Controls shall permit easy selection of speeds from minimum to maximum speeds when handling the rated load and also to meter air supply to motors for vernier positioning of the load vertically and horizontally when tested as specified in 4.5.28.1 in very limited space. “UP” and “DOWN” controls shall be labeled as such. Traversing controls shall be color-coded to indicate the direction of travel, and corresponding colored arrows shall be affixed to the hoist and shall be visible from operating positions on each side of the hoist.

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3.6.24.1 Shuttle valves. Shuttle valves or similar devices shall be provided. In the event the handcrank for manual operation (see 3.6.25) is removed from the hoist or when non-overhauling manual transmission (see 3.6.25.1) is engaged, the pendant unit (see 3.6.24) shall not release brake pressure on the hoist or tractor motor. The motor shall resist back drive by the rated load at the brake release pressure. The brake release to drive time-lag shall not result in any perceptible slippage of lift or traverse motion.

3.6.24.2 Dump valves. Pendants that are longer than the minimum required length to operate the system shall be provided with dump valves or similar devices to meet the vernier positioning requirements of 4.5.28.1.

3.6.25 Manual operation. Means shall be provided for manual lowering and traversing operation in the event of a loss of air supply. To provide hand clearance for operator safety, the length and location of the handcrank shall provide for a minimum of 1 inch of operational hand clearance measured vertically between the handcrank and the top of the smallest specified I-beam trolley track, track foundation, or hull structure. The force required on a crank to lower the rated load shall be not greater than 40 pounds. The load shall not lower unless the brakes are intentionally and manually released (see 3.6.23) or the handcrank is manually cranked. Means shall be provided so that power operation shall not be possible when the handcrank is removed from its stowage position.

3.6.25.1 Non-overhauling manual transmissions. When specified (see 6.2), worm gear type non-overhauling transmissions for emergency manual lifting, lowering, and traversing of the load shall be provided. When engaged, the load shall not overhaul the transmission when the brake is released and the handcrank is free from a restraining force. The manual brake release, hoisting, and traversing mechanisms shall be arranged so they can be operated by the operator standing on the deck without endangering the operator. The force required on a handcrank to raise, lower, or traverse a load shall be not greater than 40 pounds.

3.6.26 Air supply. The air supply line shall connect to the hoist (not to the pendant control).

3.6.27 Repair. Means shall be provided for adjustment, service, and replacement of all operating components of the trolley hoist. Parts and assemblies subject to possible periodic maintenance or overhaul shall be maintainable as specified in 3.6.33.3. Adjustment means for brake setting and air control valves shall be easily available. There shall be no interference to the servicing or draining of oils and lubricants, if applicable, to or from any assembly or component by frame members or other obstructions. Grease seals shall be provided to prevent leakage of oil and grease.

3.6.28 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.6.29 Identification and instruction plates.

3.6.29.1 Identification plates. Identification plates shall be provided in a prominent position on the side plate of the hoist and control unit. Identification plates shall be corrosion-resisting steel or aluminum alloy in accordance with MIL-DTL-15024 and MIL-P-15024/5, types A, B, C, or H. The plates shall identify the following:

a. Weight and shock (grade), as applicable
b. Rated load and hoisting speed
c. Manufacturer’s name and part, serial, or model number
d. Military specification (MIL-DTL-24591)
e. Type
f. Contract or order number
g. Date of manufacture
h. National Stock Number (NSN) (if established)

3.6.29.2 Instruction and caution plates. Operating instructions shall be clearly inscribed on the control pendant. Instruction and caution plates shall be of corrosion-resisting steel or aluminum alloy in accordance with MIL-DTL-15024 and MIL-P-15024/5, types A, B, C, or H.

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3.6.30 Lubrication. Means shall be provided for lubrication of all moving parts of the hoist. Lubrication fittings shall be immediately accessible, except for removal of housing covers for internal lubrication points. Where practicable, points requiring lubrication shall be permanently lubricated. Bearing lubrication and protection shall conform to 3.6.21. Grease and oil lubrication requirements shall be in accordance with MIL-PRF-17331. Other lubricants are acceptable if included in the Navy stock system and free of ozone depleting substances.

3.6.31 Painting. Exposed surfaces, except hooks, chains, and other wear surfaces, shall be cleaned, pretreated, and finish coated in accordance with the requirements herein. Paints and coatings shall be lead free and chromate free. Exposed surfaces, except hooks, chains, and other wear surfaces, shall be painted as specified herein. Prepare surfaces to be painted by abrasive blasting to SSPC-SP 10 near-white metal condition and establish a surface profile of 2 to 4 mils; if abrasive blasting to SSPC-SP 10 is not possible, then new bare steel housing shall be chemically cleaned and pre-treated with a zinc or iron phosphate treatment per TT-C-490. The first coat of paint applied shall be MIL-DTL-24441, type III, formula 159 zinc-rich epoxy at a thickness of 3 to 4 mils dry film thickness (DFT). The second coat of paint applied shall be MIL-DTL-24441, type IV, formula 150 green epoxy at a thickness of 4 to 6 mils DFT. The top coat of paint, with the color as specified (see 6.2), applied shall be a MIL-PRF-24635, type V or VI polysiloxane coating at a thickness of 5 to 8 mils DFT. All manufacturer's instructions shall be followed regarding application conditions and procedures (e.g., temperatures, humidity, pot life, cure times, etc.).

3.6.32 Operating instructions. The operating instructions manual shall be placed in a waterproof envelope marked "Operating Instructions". The envelope shall be attached securely to the hoist.

3.6.33 Availability, reliability, and maintainability.

3.6.33.1 Availability. The minimum acceptable inherent availability (Ai) of the hoist shall be 0.90. This requirement establishes threshold values for reliability, maintainability, and supportability of the hoist. Availability is defined in MIL-HDBK-470.

3.6.33.2 Reliability. The hoist shall operate for an average period of 3,000 continuous cycles without failure. This value of 3,000 mean cycles between failure (MCBF) is equivalent to 90 days of normal ship operation without hoist failure. Reliability is defined in MIL-HDBK-781.

3.6.33.3 Maintainability. Routine corrective maintenance at the organizational level shall be accomplished by replacing complete assemblies and subassemblies. The mean time to repair (MTTR) for the hoist shall be 4 hours. Not less than 95 percent of all corrective maintenance actions shall require not more than 10 hours to complete. Maintainability is defined in MIL-HDBK-781.

3.6.34 Workmanship. The hoist shall withstand any operational requirements specified herein without permanent deformation, breakage, malfunction, or component interference caused by improper workmanship. Parts of the hoist, before and after painting, shall be clean and free of sand, rust, dirt, fins, pits, sprues, scales, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded so that a hazardous surface does not exist. Bolted connections shall use standard bolts and nuts; self-locking nuts are acceptable. Threads shall have full thread engagement.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on pneumatic, trolley-type chain hoists in accordance with table II when a first article sample is required (see 3.1). One first article sample successfully examined and tested for acceptability of operating characteristics as specified in 4.4 and 4.5 shall serve as a manufacturing standard and be used as a guide in the inspection of production hoists produced under the contract. If required (see 6.2), one other first article model shall be tested in accordance with MIL-DTL-901 (see 3.6.8 and 4.5.12).

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TABLE II. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Materials	X	X	3.2	4.5.1
Safety criteria	X	--	3.3	4.5.2
Performance and characteristics	X	X	3.4	4.5.3
Cycling test	X	X	3.4	4.5.3.1
Lifecycle test	X	--	--	4.5.3.2
Incline consideration	X	--	3.5	4.5.4
Weight of system	X	--	3.6.1	4.5.5
Strength	X	--	3.6.2	4.5.6
Moderate sea conditions	X	--	3.6.3	4.5.7
Temperature	X	--	3.6.4	4.5.8
Service life	X	--	3.6.5	4.5.9
Fail-safe operation	X	X	3.6.6	4.5.10
Stress	X	--	3.6.7	4.5.11
Loading factor	X	X	3.6.7.1	4.5.11.1 and 4.5.11.2
Shock	X	--	3.6.8	4.5.12
Mechanical vibration	X	--	3.6.9	4.5.13
Noise	X	X	3.6.10	4.5.14
Nuclear lifting	X	X	3.6.11	4.5.15
Construction	X	X	3.6.12	4.5.16
Rotating shafts	X	X	3.6.13	4.5.17
Frame or housing	X	X	3.6.14	4.5.18
Gears	X	--	3.6.15	4.5.19
Hoist drives	X	--	3.6.16	4.5.20
Hoist lift	X	--	3.6.16.1	4.5.20.1
Hoist traverse unit drive	X	X	3.6.16.2	4.5.20.2
Load chain	X	X	3.6.17	4.5.21
Load chain finish	X	X	3.6.17.1	4.5.21.1
Load chain stowage	X	X	3.6.17.2	4.5.21.2
Load chain wheel, sprocket, and shaft	X	X	3.6.17.3	4.5.21.3
Chain guides	X	X	3.6.17.4	4.5.21.4
Chain replacement	X	--	3.6.17.5	4.5.21.5
Limit stops	X	X	3.6.18	4.5.22
Load hooks	X	X	3.6.19.1	4.5.23.1
Load hook finish	X	X	3.6.19.1.1	4.5.23.1.1
Shackle	X	X	3.6.19.2	4.5.23.2
Range of load hook	X	X	3.6.20	4.5.24
Bearings	X	--	3.6.21	4.5.25

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TABLE II. First article and conformance inspections - Continued.

Characteristic	First article	Conformance	Requirement	Verification
Trolley	X	X	3.6.22	4.5.26
Hoist brakes	X	X	3.6.23	4.5.27
Pendant control	X	X	3.6.24	4.5.28 and 4.5.28.1
Shuttle valve	X	X	3.6.24.1	4.5.28.2
Dump valve	X	X	3.6.24.2	4.5.28.3
Manual operation	X	X	3.6.25	4.5.29
Non-overhauling manual transmissions	X	X	3.6.25.1	4.5.29.1
Air supply	X	X	3.6.26	4.5.30
Repair	X	--	3.6.27	4.5.31
Interchangeability	X	--	3.6.28	4.5.32
Identification and instruction plates	X	X	3.6.29	4.5.33
Lubrication	X	--	3.6.30	4.5.34
Painting	X	X	3.6.31	4.5.35
Operating instructions	X	X	3.6.32	4.5.36
Availability	X	--	3.6.33.1	4.5.37.1
Reliability	X	--	3.6.33.2	4.5.37.2
Maintainability	X	--	3.6.33.3	4.5.37.3
Workmanship	X	X	3.6.34	4.5.38

4.3 Conformance inspection. Conformance inspection shall be performed in accordance with table II on samples selected in accordance with 4.3.2 and 4.3.3.

4.3.1 Inspection lot. For the purposes of conformance inspection and test sampling, a lot is defined as all the hoists of the same size and type, produced in one facility, using the same production processes and materials, and being offered for delivery at one time. The sample unit shall be one complete hoist unit.

4.3.2 Sampling for visual examination. As a minimum, a sample quantity shall be randomly selected from each lot of completed hoists in accordance with table III and examined in accordance with 4.5 for the defects listed in table V. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, 100 percent of the lot may be screened for the defective characteristic(s) or a new lot may be provided, which shall be examined in accordance with the sampling plan contained herein. Requirements for maintenance of records of inspections, tests, and any resulting rejections shall be as specified (see 6.2).

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TABLE III. Sampling for visual examination.

Lot size	Sample size	
	Major defects	Minor defects
2 to 25	5	3
26 to 50	5	5
51 to 90	7	6
91 to 150	11	7
151 to 280	13	10
281 to 500	16	11
501 to 1,200	19	15
1,201 to 3,200	23	18

4.3.3 Sampling for tests. As a minimum, a sample quantity shall be randomly selected from each lot of completed hoists in accordance with table IV and subjected to the tests specified in 4.5. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, 100 percent of the lot may be screened for the defective characteristic(s) or a new lot may be provided, which shall be tested in accordance with the sampling plan contained herein. Requirements for maintenance of records of inspections, tests, and any resulting rejections shall be as specified (see 6.2).

TABLE IV. Sampling for tests.

Lot size	Sample size
2 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1,200	19
1,201 to 3,200	23

4.4 Examination. Sample hoist units selected as specified in 4.3.2 shall be assessed to verify conformance to requirements. For sample hoist components that require visual examination, the visual defects shall be classified as specified in table V.

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TABLE V. Classification of visual defects.

Category	Defect
Major	
101	Materials not as specified (see 3.2).
102	Hoist clearance, underside of track to inside of load hook, not as specified (see 3.4.1, 3.4.2, and 3.4.3).
103	Hoist construction not as specified (see 3.6.1, 3.6.4, 3.6.11, 3.6.12, 3.6.13, 3.6.15, and 3.6.17.5).
104	Gear casing does not protect against dirt, dust, and water spray or permit ready access (see 3.6.15).
105	Chain links not uniform size (see 3.6.17).
106	Hoist lift range not as specified (see 3.4.1, 3.4.2, and 3.4.3).
107	Frame or housing not as specified (see 3.6.14).
108	Hoist drives not as specified (see 3.6.16).
109	Load chain wheel, sprocket, and shaft not as specified (see 3.6.17.3).
110	Load chain not securely attached to hoist or not easily removed as specified (see 3.6.17).
111	Load chain reel, bag, or basket does not store full length of chain (see 3.6.17.2).
112	Chain reel does not maintain tension on chain as specified (see 3.6.17.2).
113	Hoist hooks not equipped with safety devices, or safety devices not as specified (see 3.6.19.1).
114	Exposed bearings not sealed or shielded (see 3.6.21).
115	Pendant unit not marked, of the required length for the lift range, or able to support a 200-pound load, as specified (see 3.6.24).
116	Handcrank clearance not as specified (see 3.6.25).
117	Hoist workmanship not as specified (see 3.6.34).
118	Hoist components damaged or missing.
Minor	
201	Hoist weight not as specified (see 3.4.1, 3.4.2, and 3.4.3).
202	Hoist length and width dimensions not as specified (see 3.4.1, 3.4.2, and 3.4.3).
203	Load chain finish not as specified (see 3.6.17.1).
204	Chain guides do not completely enclose the chain as specified (see 3.6.17.4).
205	Shackle not present, if specified, or shackle does not have correct opening size or rotate as specified (see 3.6.19.2).
206	Trolley wheel mounting arrangement not as specified (see 3.6.22).
207	Air supply line not connected to hoist as specified (see 3.6.26).
208	Hoist identification plate missing or not attached as specified (see 3.6.29.1).
209	Instruction and caution plates not as specified (see 3.6.29.2).
210	Lubrication fittings missing or damaged (see 3.6.30).
211	Lubrication of moving parts not as specified (see 3.6.30).
212	Paint coating not as specified (see 3.6.31).

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4.5 Design verification. Sample units selected in accordance with 4.2 and 4.3 shall be tested as specified in 4.5.1 through 4.5.38 and table II. Hoists shall operate smoothly without jerks, jars, or excessive vibration during testing. In no case shall a load hook slip during operation or from a stopped position.

4.5.1 Materials. Materials shall be visually inspected and tested to determine compliance with 3.2 and its sub-sections. Conformance inspection shall be limited to checking that materials are in accordance with 3.2 through 3.2.4.

4.5.1.1 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.2 and 6.2).

4.5.2 Safety criteria. The hoist shall be visually inspected and operated to determine compliance with 3.3.

4.5.3 Performance and characteristics. The hoist shall be visually inspected and tested to determine compliance with 3.4.1, 3.4.2, and 3.4.3 for type I, type II, and type III hoists, respectively. Each installed hoist shall be subjected to a cycling test in accordance with 4.5.3.1 and a lifecycle test, as specified (see 6.2), in accordance with 4.5.3.2.

4.5.3.1 Cycling tests. With the hoist installed on a rail that meets the radius requirements specified in 3.4.1, 3.4.2, and 3.4.3, the hoist shall be operated through 50 continuous operating cycles. The rail shall include a section of track of the radius specified in 3.4.1, 3.4.2, and 3.4.3, as applicable. During the test, the hoist shall lift and traverse the maximum rated load. The load shall be accelerated to the maximum rated speed specified in 3.4.1, 3.4.2, and 3.4.3. A cycle shall consist of raising the load for a 10-second period, stopping and traversing for a 10-second period, stopping and re-traversing for a 10-second period, and stopping and finally lowering for a 10-second period to return to the original starting point, followed by a 40-second rest period, after which the test shall be repeated. During each 10-second operating period, the load shall be smoothly accelerated from the stopped position to maximum speed and back to the stopped position. The hoist shall then be continuously operated at maximum lifting and lowering speeds through approximately 90 percent of the design lift height for not less than 30 minutes.

4.5.3.2 Lifecycle test. When specified (see 6.2), one first article sample hoist (see 4.2) shall be run through 500 consecutive cycles free of malfunction. Operation time, cycle counts, description of preventative maintenance action, and description of malfunctions and resulting repair actions shall be logged.

4.5.4 Incline consideration. The hoist shall be tested to verify compliance with 3.5 and the following: The load hook carrying the rated load shall be raised and lowered with the hoist alternately on a ±10-degree incline parallel to the rail and with the hoist tilted 10 degrees to either side of the rail. There shall be no traverse movement along the rail, no binding of the drive or jamming of the load chain in the guides, and no degradation of any operating function specified herein. After this test, the hoist shall be operated on a horizontal track without evidence of binding or inability to perform the required functions. With the hoist mounted on a 15-degree incline rail, a rated load shall be suspended from the hook for a period of 2 minutes without any hoist movement along the rail. The hoist shall be traversed at reduced speed up the 10-degree incline with the rated load suspended from the load hook through the full lift range.

4.5.5 Weight of system. The weight of the system and analysis of the maintenance requirements shall be provided to verify compliance with 3.6.1.

4.5.6 Strength. Calculations shall be performed and measurements taken to verify compliance with 3.6.2.

4.5.7 Moderate sea conditions. A Finite Element Analysis (FEA), computer modeling and simulation, or calculations shall be performed to show that the hoist system or bridge crane can operate with the added dynamic forces of operating in moderate sea conditions to determine compliance with 3.6.3.

4.5.8 Temperature. Operational testing and visual inspection shall be conducted to verify compliance with 3.6.4. Endurance testing shall be conducted as follows to verify compliance with 3.6.4: One sample hoist of each class and type shall be subjected to 3,000 cycles of continuous operation. In the event a failure occurs during this test, and an investigation and design study result in modification to the test unit and remaining units as specified (see 6.2), the endurance test shall be repeated. After completion of the endurance tests, the gears, chain, bearings, chain sprockets, brakes, and other wearing parts shall be examined for excessive wear. During operation of these hoists, no wear particles greater than 0.031 inch in any direction shall be generated. Life-lubricated bearings requiring lubrication, load-brake lining reduced in excess of 50 percent of useful life, and reduction of bar diameter

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of link chain in excess of 10 percent shall be considered excessive wear. The tests specified in 4.5.11.1 and 4.5.11.2 shall be repeated to demonstrate safe operation.

4.5.9 Service life. The hoist shall be analyzed and tested to verify compliance with 3.6.5.

4.5.10 Fail-safe operation. The fail-safe features shall be operationally tested to verify compliance with 3.6.6.

4.5.11 Stress. FEA, computer modeling and simulation, or calculations shall be conducted to verify compliance with 3.6.7 and 3.6.7.1. The hoist shall be subjected to load tests as specified in 4.5.11.1 and 4.5.11.2, during which the hoists shall lift and lower a load within their rated load capacity without jerks, jars, or vibration.

4.5.11.1 Static load test. With the hoist mounted on a section of rail, a test load of 2.0 times the rated load shall be suspended from the load hook for a period of 10 minutes. There shall be no permanent deformation of any portion of the hoist frame or load support or evidence of chain failure. The hoist shall not be deformed or otherwise affected in any way that would cause degradation of any operating function as a result of this test. At the end of the test period, the load shall be slowly swiveled 360 degrees not less than twice in each direction, alternately, without any binding or malfunction.

4.5.11.2 Dynamic load test. With the hoist mounted on a section of rail, a test load of 1.5 times the rated load shall be lifted and lowered through the full hoisting range not less than ten times in each direction, with the hoist alternately tilted on a 10-degree incline to either side of the rail at maximum speed or highest speed capable (if less than maximum rated speed). The load shall be stopped twice during each lowering cycle, once at mid-point and once near the lower limit. There shall be no binding of the drive or jamming of the load chain in the guides and no degradation of any operating function specified herein, except that maximum speed requirements shall not apply. There shall be no permanent deformation of any portion of the hoist frame or load support or evidence of chain failure.

4.5.12 Shock tests. When specified (see 6.2), shock tests shall be conducted in accordance with the requirements of MIL-DTL-901 for grade A equipment and the hoist shall meet the requirements of 3.6.8. Testing shall be conducted on one complete unloaded hoist of each type. Shock qualification shall be certified on the applicable hoist testing placard. In accordance with MIL-DTL-901, shock test requirements may be fulfilled if an identical hoist conforming to the requirements of this certification has been successfully shock tested and meets the requirements of 3.6.8.

4.5.13 Mechanical vibration. The hoist shall be vibration tested in accordance with MIL-STD-167-1, type I to verify compliance with 3.6.9.

4.5.14 Noise. The hoist shall be tested for noise during operation in accordance with MIL-STD-1474 to verify compliance with 3.6.10.

4.5.15 Nuclear lifting. When specified (see 6.2), analysis or calculations shall be performed to verify compliance with 3.6.11.

4.5.16 Construction. A visual inspection and testing shall be performed to verify that the hoist was constructed in compliance with 3.6.12.

4.5.17 Rotating shafts. A visual inspection shall be performed to verify compliance with 3.6.13.

4.5.18 Frame or housing. A visual inspection and analysis shall be performed to verify compliance with 3.6.14.

4.5.19 Gears. A visual inspection shall be performed to verify compliance with 3.6.15.

4.5.20 Hoist drives. Analysis, calculations, and testing shall be performed to verify compliance with 3.6.16.

4.5.20.1 Hoist lift drive. A visual inspection and test shall be performed to verify compliance with 3.6.16.1.

4.5.20.2 Hoist traverse unit drive. A visual inspection and test shall be performed to verify compliance with 3.6.16.2.

4.5.21 Load chain. A visual inspection, analysis or calculations, and testing shall be performed to determine compliance with 3.6.17.

4.5.21.1 Load chain finish. A visual inspection and analysis shall be performed to verify compliance with 3.6.17.1.

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4.5.21.2 Load chain stowage. A visual inspection and analysis or testing shall be performed to verify compliance with 3.6.17.2.

4.5.21.3 Load chain wheel, sprocket, and shaft. A visual inspection and testing shall be performed to verify compliance with 3.6.17.3.

4.5.21.4 Chain guides. A visual inspection and testing shall be performed to verify the chain guide design as specified in 3.6.17.4.

4.5.21.5 Chain replacement. A visual inspection and testing shall be performed to verify compliance with 3.6.17.5.

4.5.22 Limit stops. The limit stops shall be visually inspected to verify compliance with 3.6.18 and shall be tested in accordance with 4.5.24.

4.5.23 Load attachments. The load attachments shall be verified in accordance with the following paragraphs.

4.5.23.1 Load hooks. A visual inspection and testing shall be performed to verify compliance with 3.6.19.1. All load hooks shall be tested as specified in 4.5.24 to verify compliance with 3.6.20.

4.5.23.1.1 Load hook finish. A visual inspection and analysis shall be performed to verify compliance with 3.6.19.1.1.

4.5.23.2 Shackle. A visual inspection and testing shall be performed to verify compliance with 3.6.19.2.

4.5.24 Range of load hook. Testing shall be performed to verify compliance with 3.6.20. The hoist shall lift a rated load placed in a geometric configuration such that the angle of the load chain to the load chain sprocket would be the same as the angle resulting from picking up a load in a radius of 19 inches of a horizontal plane at 7 feet below the trolley track. The load shall be lifted not less than four times with the load being placed, for each lift, in progressive 90-degree positions of the projected plane described above. In each case, the hoist shall successfully pick up the load without jamming, binding, or fouling of the load chain or any component of the drive assembly. Upper and lower limit stops shall be verified by lowering and raising the load hook through the maximum range in 3.6.20.

4.5.25 Bearings. A visual inspection, analysis, and testing shall be performed to verify bearing compliance with 3.6.21.

4.5.26 Trolley. A visual inspection and testing shall be performed to verify compliance with 3.6.22.

4.5.27 Hoist brake. The hoist brake shall be tested to verify compliance with 3.6.23 and the traversing brake test in 4.5.27.1.

4.5.27.1 Traversing brake test. Traversing brakes shall be tested by subjecting the stopped hoist to a 1,560-pound pull that shall be exerted in each direction parallel to the trolley rail. Brakes shall keep the loaded hoist stationary on straight rails with the load applied in each direction alternately. The hoist shall then be traversed in both directions on straight sections of rail with the rated load suspended from the hook. During this traversing, the hoist shall be safely stopped from maximum speed not less than five times with no evidence of sliding. Satisfactory operation of the brakes shall be determined for 5-inch, 6-inch, or 8-inch I-beam rails. The hoist shall traverse at rated speed with the rated load suspended from the hook.

4.5.28 Pendant control. A visual inspection of the pendant control and testing in accordance with 4.5.28.1 shall be performed to verify compliance with 3.6.24.

4.5.28.1 Hoisting controls sensitivity test. At maximum rated load, the load hook shall be moved vertically to within 0.125 inch of a desired position. From this position, the load hook shall be moved approximately 0.50 inch up and stopped. The load hook shall be returned to within 0.125 inch of the original desired position. The load hook shall then be moved approximately 0.50 inch down, stopped, and then returned to within 0.125 inch of the original position. The load hook shall then be moved horizontally to within 0.25 inch of a desired position. From this position, the load hook shall be traversed approximately 1 inch and stopped. The load hook shall then be returned to within 0.25 inch of the desired position. The load hook shall then be traversed approximately 1 inch in the opposite direction, stopped, and then returned to within 0.25 inch of the original position. The above shall be repeated not less than six times to demonstrate the repetitive locating capability with these limits.

4.5.28.2 Shuttle valve. Testing shall be performed to verify compliance with 3.6.24.1.

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4.5.28.3 Dump valve. A visual inspection and testing in accordance with 4.5.28.1 shall be performed to determine compliance with 3.6.24.2.

4.5.29 Manual operation. The manual operation of the hoist shall be tested to verify compliance with 3.6.25. The hoist shall traverse (where applicable) and lift and lower through the full hoisting range with rated load, or lower only when manual brake release only is specified (see 3.6.25.1). While the hoist is being manually operated, the interlock shall prevent pneumatic operation. With the loaded hoist set up for manual operation, the hoist shall be manually traversed 3 feet in both the forward and reverse directions using the handcrank. Then, with the handcrank disengaged and removed from the hoist, the load shall be lowered to the deck by manually releasing the load brake (see 3.6.25.1); then, using the handcrank, the load shall be manually lowered by an additional amount as necessary to release the chain tension, thus freeing the load from the hook. During manual operation, it shall be demonstrated that the operator can safely control the load speed at all times. It shall also be demonstrated that operating personnel will not be endangered if the air supply is suddenly restored during manual operation.

4.5.29.1 Non-overhauling manual transmissions. When specified (see 6.2), testing shall be performed to verify compliance with 3.6.25.1. The load hook shall be raised and lowered through its entire range using the handcrank. The load shall not overhaul the drive transmission when the handcrank is free from a restraining force, nor shall the operator be endangered if the air supply is suddenly restored.

4.5.30 Air supply. A visual inspection and testing of the air supply line shall be performed to verify compliance with 3.6.26.

4.5.31 Repair. A visual inspection and analysis shall be performed to verify compliance with 3.6.27.

4.5.32 Interchangeability. A visual inspection and analysis shall be performed to verify compliance with 3.6.28.

4.5.33 Identification and instruction plates. A visual inspection shall be performed to determine compliance with the requirements for identification plates and instruction plates in 3.6.29.1 and 3.6.29.2, respectively.

4.5.34 Lubrication. A visual inspection shall be performed to verify that lubrication is in compliance with 3.6.30.

4.5.35 Painting. A visual inspection and measurement shall be conducted to determine compliance with 3.6.31.

4.5.36 Operating instructions. The operating instructions shall be visually inspected to determine compliance with 3.6.32.

4.5.37 Availability, reliability, and maintainability.

4.5.37.1 Availability. The availability of the system shall be tested to determine the compliance with 3.6.33.1.

4.5.37.2 Reliability. The reliability shall be tested to determine the compliance with 3.6.33.2 using the guidance provided in MIL-HDBK-781.

4.5.37.3 Maintainability. The MTTR requirement (see 3.6.33.3) shall be demonstrated by the successful completion of a maintainability demonstration using the guidance provided in MIL-HBDK-781.

4.5.38 Workmanship. Construction workmanship shall be visually inspected to determine compliance with 3.6.34.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pneumatic, trolley-type chain hoists with traverse drive covered by this specification are intended for use in areas with low headroom and load eccentricity. These hoists have design and testing requirements to meet sea state conditions for operation and corrosion prevention. The pneumatic hoists may be used to handle ammunition and explosives, when certified, cargo, boats, stores, and other shipboard items vertically within a space or between decks. Current stock of pneumatic wire rope or chain hoists procured to previous revisions of this specification may be used until depleted; future requisitions for pneumatic wire rope or chain hoists should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. When first article inspection is required (see 3.1).

d. When non-standard materials are used (see 3.2.1).

e. Requirements for prohibited materials verification (see 3.2.2 and 4.5.1.1).

f. Applications for which pressed steel may be used (see 3.2.2.d).

g. Applications for which brittle material may be used (see 3.2.2.f).

h. Length of lift (range) required, if other than specified, and operating height of hoist above deck (see 3.4.1, 3.4.2, 3.4.3, 3.6.24, 4.5.3, and 4.5.28).

i. Sea state conditions at which the hoist is required to operate (see 3.6.3 and 4.5.7).

j. If construction for shock is required (see 3.6.8, 4.2, and 4.5.12).

k. If nuclear lifting is to be performed with the hoist (see 3.6.11 and 4.5.15).

l. When chain other than 1-inch pitch is required (see 3.6.16.2).

m. Whether chain reels, bags, or baskets are required (see 3.6.17.2).

n. Type of load hook required (see 3.6.19.1).

o. Load hook finish, if other than specified (see 3.6.19.1.1).

p. If a shackle is required (see 3.6.19.2).

q. If non-overhauling transmission is required (see 3.6.25.1 and 4.5.29.1).

r. Color of the top coat of paint (see 3.6.31).

s. If more than one first article sample is required (see 4.2).

t. Requirements for record keeping (see 4.3.2, 4.3.3, and 6.6).

u. If lifecycle testing is required (see 4.5.3.2).

v. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be preproduction sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. The contracting officer should also include specific Technical Authority approval of first article tests and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1). Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is cited on a contract. If technical manuals are required, they must be acquired under a separate contract line item in the contract.

6.5 Shock tests. When an identical hoist conforming to the requirements of this specification has been successfully shock tested, the test report may be submitted to the contracting activity as fulfillment of the requirements for shock test.

6.6 Record keeping. Records of inspections, tests, and any resulting rejections should be maintained for a period of 3 years after contract completion.

6.7 Definitions.

6.7.1 Brittle material. Material showing less than 10 percent elongation in 2 inches for the standard tensile test. For the static loading case (normal ship operation), brittle material is material used below its nil ductility transition (NDT) temperature as measured by the Naval Research Laboratory (NRL) drop weight test. Charpy V-notch impact strength values may be used as criteria when these have been correlated with drop weight test results.

6.8 Subject term (key word) listing.

Airborne noise
Load hook
Pendant unit
Reliability and maintainability
Shackle
Shuttle valve

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 3950-2023-001)

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